Descriptive Statistics Project

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Project: The number of Larceny thefts committed in Queen Anne's County from 1975-2020.

The sample size is 46

URL: [Violent_Crime___Property_Crime_by_County__1975_to_Present.xlsx]

Violent_Crime__Prope

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	A	B
1	YEAR	LARCENY THEFT
2	1975	356
3	1976	335
4	1977	294
5	1978	327
6	1979	364
7	1980	432
8	1981	467
9	1982	424
10	1983	438
11	1984	417
12	1985	458
13	1986	405
14	1987	402
15	1988	506
16	1989	479
17	1990	567
18	1991	545
19	1992	551
20	1993	629
21	1994	544
22	1995	562
23	1996	643
24	1997	611
25	1998	660
26	1999	776
27	2000	587
28	2001	552
29	2002	615
30	2003	644
31	2004	668
32	2005	662
33	2006	674
34	2007	676
35	2008	685
36	2009	681
37	2010	741
38	2011	700
39	2012	623
40	2013	576
41	2014	540
42	2015	490
43	2016	456
44	2017	459
45	2018	352
46	2019	334
47	2020	267

$356, 335, 294, 327, 364, 432, 467, 424, 438, 417, 458, 405, 402, 506, 479, 567, 545,$
$551, 629, 544, 562, 643, 611, 660, 776, 587, 552, 615, 644, 668, 662, 674, 676,$
$685, 681, 741, 700, 623, 576, 540, 490, 456, 459, 352, 334, 267$

In this project I shall:

(1). Present the data in the form of a graph.

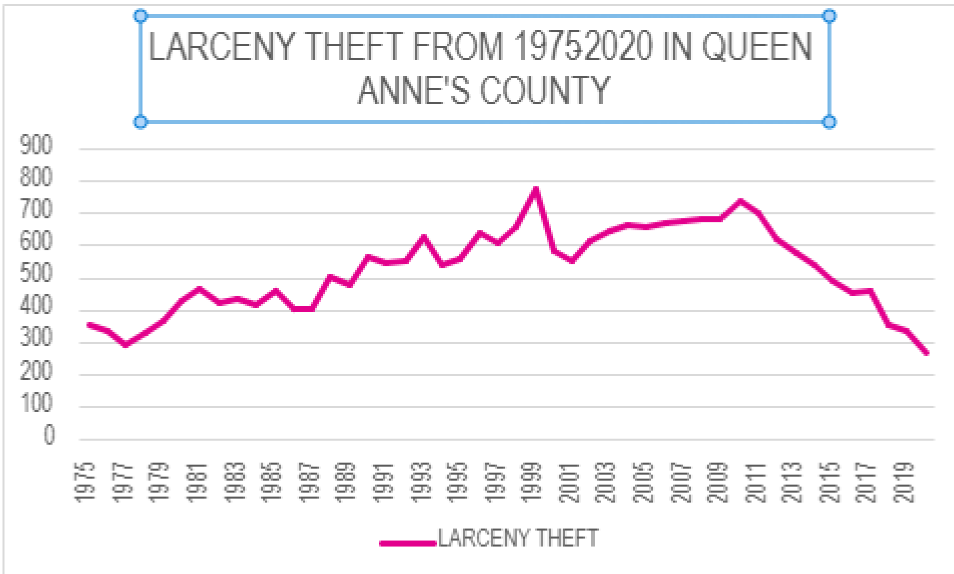
(2). Calculate the measures of center, such as mean, median, mode and midrange.

(3). Calculate the measures of spread, such as range, variance and standard deviation, and interquartile range.

(4). Calculate the measures of position, such as the five-number summary of the data.

Description: Larceny Theft in Queen Anne's County from 1975-2020

Data Presentation:



The highest amount of larceny thefts was recorded in 1999 and it stayed at 776 until its steady decline starting in 2011 and continuing into 2020.

Let x be the number of larceny thefts committed from 1975-2020 in Queen Anne's County.

Data Analysis:

	A	B
1	*LARCENY THEFT*	
2		
3	Mean	525.5217391
4	Standard Error	19.3246542
5	Median	544.5
6	Mode	#N/A
7	Standard Deviation	131.0661816
8	Sample Variance	17178.34396
9	Kurtosis	-0.97814112
10	Skewness	-0.143609435
11	Range	509
12	Minimum	267
13	Maximum	776
14	Sum	24174
15	Count	46

Measures of Center:

Mean:

$$\begin{aligned}\Sigma x = \ & 356 + 335 + 294 + 327 + 364 + 432 + 467 + 424 + 438 \\ & + 417 + 458 + 405 + 402 + 506 + 479 + 567 + 545 \\ & + 551 + 629 + 544 + 562 + 643 + 611 + 660 + 776 \\ & + 587 + 552 + 615 + 644 + 668 + 662 + 674 + 676 \\ & + 685 + 681 + 741 + 700 + 623 + 576 + 540 + 490 \\ & + 456 + 459 + 352 + 334 + 267 = 24{,}174\end{aligned}$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{24174}{46} = 525.5217391 \cong 526$$

The mean average of larceny thefts in the county of Queen Anne's between the years of 1975-2020 is approximately 526 thefts.

Median:

I will now put the data into ascending order.

$267, 294, 327, 334, 335, 352, 356, 364, 402, 405, 417, 424, 432, 438, 456, 458, 459, 467,$

$479, 490, 506, 540, 544, 545, 551, 552, 562, 567, 576, 587, 611, 615, 623, 629, 643, 644,$

$660, 662, 668, 674, 676, 681, 685, 700, 741, 776$

$$Median = Q_2 = \frac{544 + 545}{2} = \frac{1{,}089}{2} = 544.5 \cong 545$$

When the number of larceny thefts in Queen Anne's county is put in ascending order we can see that the median is approximately 545 thefts.

Mode:

There is no number in this data set that appears more than once.

Midrange:

The minimum amount of larceny thefts is 267 and the maximum number of thefts is 776.

To find the midrange we must add the minimum and the maximum,

$$Midrange = \frac{minimum + maximum}{2} = \frac{267 + 776}{2} = 521.5 \cong 522$$

So we can see that between 1975-2020 the average of the minimum amount of larceny thefts in the county and the maximum amount add up to 522 thefts.

Measures of Spread:

$Range = \ maximum - minimum$

$776 - 267 = 509$

The difference between the maximum amount of larceny's and the minimum amount in the county of Queen Anne's is 509 thefts.

Variance:

Number of Larceny Thefts. x	$x - \bar{x}$	$(x - \bar{x})^2$
356	-169.5217391	28737.62004
335	-190.5217391	36298.53308
294	-231.5217391	53602.31569
327	-198.5217391	39410.88091
364	-161.5217391	26089.27221
432	-93.52173913	8746.31569
467	-58.52173913	3424.793951

424	−101.5217391	10306.66352
438	−87.52173913	7660.05482
417	−108.5217391	11776.96786
458	−67.52173913	4559.185255
405	−120.5217391	14525.4896
402	−123.5217391	15257.62004
506	−19.52173913	381.0982987
479	−46.52173913	2164.272212
567	41.47826087	1720.446125
545	19.47826087	379.4026465
551	25.47826087	649.1417769
629	103.4782609	10707.75047
544	18.47826087	341.4461248
562	36.47826087	1330.663516
643	117.4782609	13801.14178
611	85.47826087	7306.533081
660	134.4782609	18084.40265
776	250.4782609	62739.35917
587	61.47826087	3779.57656

552	26.47826087	701.0982987
615	89.47826087	8006.359168
644	118.4782609	14037.0983
668	142.4782609	20300.05482
662	136.4782609	18626.31569
674	148.4782609	22045.79395
676	150.4782609	22643.70699
685	159.4782609	25433.31569
681	155.4782609	24173.4896
741	215.4782609	46430.88091
700	174.4782609	30442.66352
623	97.47826087	9502.011342
576	50.47826087	2548.05482
540	14.47826087	209.6200378
490	−35.52173913	1261.793951
456	−69.52173913	4833.272212
459	−66.52173913	4425.141777
352	−173.5217391	30109.79395
334	−191.5217391	36680.57656

$$\sum(x-\bar{x})^2 = 773025.4783$$

$$Variance, s^2 = \frac{\sum(x-\bar{x})^2}{n-1} = \frac{773025.4783}{46-1} = 17178.34396$$

The next thing we have to do is calculate the standard deviation. In order to do this we must find the square root of the variance.

$$Standard\ Deviation, s = \sqrt{variance}$$

$$= \sqrt{17178.34396} = 131.0661816$$

The standard deviation shows that there is a lot of spread within the data.

Interquartile Range:

Now we will calculate the interquartile range, which is the difference between the upper quartile and lower quartile.

$$\frac{1}{4} * 46 = 11.5$$

Round this up to 12

The 12th position in the data is 424

$$Lower\ quartile, Q_1 = 424$$

$$\frac{3}{4} * 46 = 34.5$$

Round this up to 35

The 35th position in the data is 643

$Upper\ quartile, Q_3\ =\ 643$

$Interquartile, IQR\ =\ Q_3 - Q_1$

$643 - 424 = 219$

This data shows that the variation in 50% of the larcenies in Queen Anne's County from 1975-2020 is 219 larcenies.

Measures of Position:

The five number summary of the dataset is as follows,

$Minimum$: 267

267 larcenies is the minimum number of larcenies committed in Queen Anne's County from 1975-2020.

$Lower\ Quartile$: 424

424 larcenies is the 25^{th} percentile of the data from 1975-2020 in Queen Anne's County.

$Middle\ Quartile, Median$: 544.5

Approximately 545 larcenies is the 50^{th} percentile of the data from 1975-2020 in Queen Anne's County.

$Upper\ Quartile$: 643

643 larcenies is the 75th percentile of the data from 1975-2020 in Queen Anne's County.

$Maximum$: 776

776 larcenies is the maximum number of larcenies committed in Queen Anne's County from 1975-2020.

Citations: (MLA 9)

Chukwuemeka, Samuel. "Descriptive Statistics Projects." *Descriptive Statistics*, conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html#studentsProjects. Accessed 1 Dec. 2025.

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